

HEFEIFEI JIANG

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RIVERSIDE, CA

EDUCATION

University of California, Riverside (UCR)

M.S. | *Robotics*

Sep 2025 - Dec 2026 (Expected)

Riverside, CA, USA

Shaanxi University of Science and Technology (SUST)

B.Eng. (First-class degree) | *Computer Science and Technology*

Sep 2020 - Jun 2024

Xi'an, China

Joint-Degree Undergraduate Program with Ulster University, UK

RESEARCH INTEREST

My research centers on **human-robot interaction and collaboration**: how robots and people can work together effectively in shared physical spaces. I am interested in robots that keep humans in the loop: adapting to human intent and individual preferences, learning from human demonstrations, and taking sparse guidance and corrections from people. I care about making such interaction natural, safe, and low-effort for people. I approach these problems through **robot learning**, combining imitation learning, reinforcement learning, and vision-language-action models, and deploy them on real robots, from mobile bases to manipulators. My goal is to build interactive robots that people find intuitive to guide, trust, and collaborate with.

PUBLICATIONS & PREPRINTS

* Equal contribution.

[6] Mingxuan Yan, Litian Gong, Dihong Huang, Zhixu Li, Tianyu Zhang, Wenqian Zhang, **Hefeifei Jiang**, Peihao Li, Jian Zhang, Mengfei Zhao, Yikai Tang, Hai Zhai, Zehao Wang, Ruijian Liang, Yanjia Huang, Lin Shao, Changliu Liu, Kaiyu Hang, Zhiwen Fan, Masayoshi Tomizuka, Jianfei Yang, Jiachen Li, “What Matters in Quality-Aware VLA Pretraining: A Benchmark and Empirical Study”, *IEEE International Conference on Robotics and Automation (ICRA)*, 2027. **Under review**.

[5] Shiting Gong, Jianpeng Yao, Litian Gong, **Hefeifei Jiang**, Huaide Jiang, Jinfeng Wang, Zheyu Lin, Mingyu Ding, Jiachen Li, “Closed-Loop Customizable Human Following via Distilling Monotonic Constrained Policies”, *IEEE International Conference on Robotics and Automation (ICRA)*, 2027. **Under review**.

[4] Zhefei Gong*, **Hefeifei Jiang***, Zehao Wang, Jiachen Li, “HiRe-MoMa: Hierarchical Residual Policy Learning for Real-World Mobile Manipulation with Sparse Human Guidance”, *Conference on Robot Learning (CoRL)*, 2026.

[3] Zehao Wang*, **Hefeifei Jiang***, Siyan Li, Guoyuan Wu, Jiachen Li, “CoopLight: Cooperative Vision-Language Agents for Long-Tail Traffic Control”, *International Conference on Learning Representations (ICLR)*, 2027. **Under review**.

[2] Y. Qi, M. Qiu, **H. Jiang**, F. Wang, “Extracting Fingerprint Features Using Autoencoder Networks for Gender Classification”, *Applied Sciences*, 12(19): 10152, 2022.

[1] Y. Qi, **H. Jiang**, S. Li, J. Cao, “ConvLSTM Coupled Economics Indicators Quantitative Trading Decision Model”, *Symmetry*, 14(9): 1896, 2022.

PATENTS

[2] Y. Qi, **H. Jiang**, F. Wang, J. Yang, “A Method and Apparatus for Matching a Three-dimensional Model of the Sacrum Based on Deep Learning”, Chinese patent.

[1] Y. Qi, **H. Jiang**, X. Jin, J. Ma, M. Zhao, Y. Liu, K. Qian, Y. Wang, Z. Wang, H. Cui, “AIOT Digital Twin Intelligent Substation Cloud-based Analysis Platform”, Chinese patent.

RESEARCH EXPERIENCE

Trustworthy Autonomous Systems Lab (TASL)

Advisor: Prof. Jiachen Li, UC Riverside

Oct 2025 - Present

Riverside, CA, USA

- **AXIS-Bench: Quality-Aware VLA Pretraining from Suboptimal Human Demonstrations**
 - ▶ Human demonstrations for VLA pretraining are naturally suboptimal, with hesitations and unnecessary detours.
 - ▶ AXIS-Bench is a simulation and real-world benchmark for comparing filtering, weighting, and conditioning.
 - ▶ Approach-aware episode conditioning outperforms plain behavioral cloning for all evaluated VLAs, and simulation pretraining raises real-world $\pi_{0.5}$ success from 9.6% to 31.7% on a Franka FR3.
- **Closed-Loop Customizable Human Following via Distilling Monotonic Constrained Policies**

- ▶ Preferred following distance varies across users and tasks, yet standard RL policies cannot adjust it at test time.
- ▶ The framework distills monotonic constrained-RL policies into one preference-conditioned meta-policy, with a closed-loop mapping that corrects distance drift online.
- ▶ Users can vary the following distance by up to 114% at test time, cutting collision rate by 29.68 points over planning baselines, and the policy transfers to a real ROSMASTER X3.
- **HiRe-MoMa: Real-World Fine-Tuning of Mobile Manipulation with Sparse Human Guidance**
 - ▶ Real-world fine-tuning of mobile manipulation is hard due to high-dimensional actions and base-arm coupling.
 - ▶ HiRe-MoMa learns additive residual corrections on top of a frozen base policy, decoupled into mobile-base and arm heads, and uses sparse human interventions to shape the critic through expert-guided value alignment.
 - ▶ On a TIAGo++, placement success rises from 50% to 95% with about 30 min of online interaction and <20% human intervention, regardless of the base policy architecture.
- **CoopLight: Cooperative Vision-Language Agents for Long-Tail Traffic Control**
 - ▶ Traffic signals and connected vehicles are usually controlled separately, which fails in long-tail events.
 - ▶ CoopLight is a vision-language framework in which signal and vehicle agents answer agent-specific VQA over multi-view images and exchange semantic reports for cooperative decisions.
 - ▶ In closed-loop SUMO and Blender simulation of two real intersections, CoopLight reduces average waiting time by 26% overall and 56% for connected vehicles.

Quantum Neural Networks for Efficient Multimodal Representation Learning

Jun 2024 - Dec 2024

Remote Research Intern, Advisor: Prof. Juan Zhuang

Remote

- CLIP's image encoder is large and computationally expensive, limiting the efficiency of multimodal learning.
- Quantum Neural Networks can model complex features with far fewer parameters, offering a more efficient alternative.
- A Quantum Neural Network replaces the CLIP image encoder with competitive accuracy and fewer parameters.

Summer Workshop at National University of Singapore

Jun 2024 - Jul 2024

Team Leader, Advisor: Prof. Tan Wee Kek

Singapore

- The greenhouse monitoring system combines Raspberry Pi, ESP32, and multi-modal sensors with a real-time web app.
- YOLOv8 counts plants and a ResNet classifier recognizes pests and diseases on live video streams.

Digital Twin Intelligent Grid Physical Fusion Platform

Apr 2021

Robot Embedded Development

China

- Substation inspection must run around the clock, and manual inspection exposes workers to electrical safety risks.
- A tracked inspection robot built on a Jetson Nano patrols autonomously with SLAM mapping and obstacle detection.
- A YOLOv3 model detects equipment anomalies and reads meters in real time, reporting to a companion IoT platform.
- The robot is deployed in multiple substations as a provincial-level innovation project.

SKILLS

Programming & Tools: Python, PyTorch, ROS, Git, LaTeX

Learning Methods: Imitation Learning, Reinforcement Learning, Vision-Language(-Action) Models

Simulation, Benchmarks & Datasets: IsaacLab, LIBERO, SUMO, Blender, DROID

Robots: TurtleBot3, ROSMASTER X3, TIAGo++, Franka FR3

HONORS & AWARDS

International

- **Meritorious Award**, Mathematical Contest in Modeling (MCM) Feb 2022
- **Honorable Mention**, ICPC Asia-East Continent Final Apr 2021
- **Gold Medal**, International Festival of Science and Technology, Tunisia Aug 2020

Domestic

- **"Zhi Cheng Zhi Bo" All-Round Excellence Scholarship (Top 10)**, University-level Scholarship Jun 2024
- **Zhou Lian Scholarship (Top 10, #1 by points)**, University-level Scholarship Apr 2024
- **3rd Prize**, National College Optoelectronic Design Competition Aug 2022
- **3rd Prize**, National College Computer Design Competition Aug 2022
- **Excellence Award**, China AI+ Innovation and Entrepreneurship Competition Dec 2021
- **Provincial-Level Approval**, China College Student Innovation and Entrepreneurship Project Nov 2021
- **1st Prize**, 23rd National Robot Championship & 12th International Humanoid Robot Olympics Oct 2021
- **3rd Prize**, National College Computer Design Competition Aug 2021